STELLA (YUXUAN) JIANG

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EDUCATION

University of California, Irvine Expected Mar. 2027
B.S. in Computer Science | GPA: 3.88/4.0 | Dean's Honor List
Coursework: Data Structures and Algorithms, Machine Learning, Database Management, Software Engineering

WORK EXPERIENCE

Software Engineer Intern, Applied AI | Alibaba Group (global public tech company) Jul. 2026 - Sep. 2026
CN | Tmall Campus digital services and retail ecosystem: 800+ universities and nearly 20M students.

- Implemented venue booking with **PostgreSQL/SQLAlchemy transactions**, idempotency, and a FIFO waitlist, validating 20 concurrent requests with **zero overselling** and preserved queue ordering under retries.
- Implemented a **transactional outbox and leader leases** for booking workflows, integrating Prometheus metrics and structured logs to maintain observable event delivery during worker restarts and leader failover.
- Built user-profile services transforming 100M raw events into **86.3M training records for ~1M users**, using batch jobs to isolate model training from request paths and preserve online profile-read performance.
- Integrated React/TypeScript frontends with Python services through a **Node.js BFF**, using typed DTOs, swappable adapters, and **API contract tests** to enable parallel development with mock services.
- Built a **Python RAG service** for campus knowledge search and an offline retrieval benchmark comparing BM25, BGE, and RRF on 6,562 queries, selecting BGE as the production retriever with **76.4% Top-1 and 91.3% Top-3 accuracy**.
- Built reliability controls with a **deterministic Fact Guard**, request fingerprinting, and SQLite-backed atomic counters to avoid quota consumption by failed or invalid outputs; validated with 371 backend/smoke tests, browser tests, and 999-post regression.

Software Engineer, LLM Workflow | FessorAI, UCI Capstone Apr. 2026 - Jun. 2026
Irvine, CA | Full-stack AI education platform for structured textbook generation and in-page editing.

- Implemented a **six-stage textbook generation workflow** using **Django REST and PostgreSQL** to persist content states and intermediate outputs, while supporting four LLM providers through provider-agnostic interfaces.
- Designed **Celery/Redis orchestration** for long-running generation tasks with dependency-aware scheduling, retries, **durable progress tracking, and failure recovery**, validating reliability through 49 focused tests and fault-injection scenarios.
- Streamed text and tool events via **Django Channels/Redis/WebSockets** into React blocks for immediate navigation and review; extended a **28-tool editing assistant** with native function calling and conversation memory.
- Fixed long-form editing by removing a **1,000-character truncation** and passing full-section context.

Software Engineer Intern | NextTier Dec. 2025 - Feb. 2026
Sacramento, CA | Backend and evaluation services for a policy-grounded expense-audit copilot.

- Built an AI expense-audit service using **LangChain-based RAG**, OpenAI embeddings, and vector search to generate policy-grounded approve/review/reject recommendations, reducing median review time from **6.4 to 2.1 min/report**.
- Designed a **FastAPI/Pydantic backend** separating stateless business services from stateful policy and receipt storage, with typed validation, error handling, structured logging, and health checks to improve reliability and observability.
- Implemented **LLM function calling** for expense API queries, cutting model calls from **2.5 to 1.6 per audit** and token cost by **36%**.
- Containerized services with Docker and built **MLflow evaluation pipelines** comparing retrieval quality, decision accuracy, latency, and token cost before releases.

Founder & Software Engineer | FocusTrail Dec. 2025 - Present
Irvine, CA | ADHD-friendly AI task execution coach supported by UCI Startup Fund; selected for Alibaba Tmall Campus OPC (5/50).

- Shipped a **local-first AI task execution architecture** with pre-inference privacy filtering that keeps raw activity data on-device.
- Implemented a Gemma-powered, **versioned task-DAG runtime** with dependency-aware replanning, plan validation, and state reconciliation to protect completed work and maintain execution consistency.
- Built React/Node.js/Express/SSE applications and **cross-platform activity agents** with task-aware classification, local/cloud model routing, provider failover, and real-time model-availability feedback.

Software Engineer Intern | Join-Cheer (public enterprise software company, 1M+ users) Jun. 2025 - Sep. 2025
CN | Enterprise AI-agent platform and data services for government and financial-reporting workflows.

- Led a **Dify-to-Nuwa GPT migration** for an internal agent platform, preserving **legacy workflow compatibility** in production.
- Built Python/Flask services on ECS/OSS to normalize structured and unstructured data into **agent-callable tools**.
- Automated **Docker/Kubernetes CI/CD** for 10+ services, cutting **deployment time by ~40%** and eliminating manual rollbacks.
- Instrumented Function Compute with health checks/runtime metrics to surface post-migration failures across distributed services.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL
Frontend: React, Next.js, WebSocket, Server-Sent Events
Backend & Systems: Distributed Systems, Microservices, REST APIs, FastAPI, Django, Flask, Node.js/Express, Celery
Databases & ORM: PostgreSQL, Redis, SQLite, SQLAlchemy
DevOps: Linux, Git, Docker, Kubernetes, CI/CD, GitHub Actions, Alibaba Cloud, AWS, Prometheus
Testing: Pytest, Unit/Integration Testing, API Contract Testing, Regression Testing
AI & ML: RAG, Information Retrieval, Natural Language Processing (NLP), AI Agents, Function Calling, Structured Outputs, Model Evaluation